Disputes & Debates: Editors' Choice

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Editors' Note: Opinion and Special Articles: Remote Evaluation of Acute Vertigo: Strategies and Technological Considerations

In "Opinion and Special Articles: Remote Evaluation of Acute Vertigo: Strategies and Technological Considerations," Green et al. discussed the telemedicine approach to a patient with acute vertigo by describing a modified version of the penlight-cover test to detect nystagmus, the alternate-cover test to check for a skew deviation and techniques to perform a virtual Dix-Hallpike maneuver and the head thrust test. Tjell agreed with the strategies described, but reported they may be challenging in a busy emergency department, and indicated a preference for 3 different assessments including the Romberg test, the horizontal smooth pursuit eye movement test, and tragal pressure. Green et al. noted that evaluation of nystagmus by smooth pursuit has pitfalls and the diagnostic value of tragal compression is unclear. They suggested that a very large multicenter international trial would be needed to clarify whether there is a difference in diagnostic accuracy between their proposed batteries or a combination of their tests.

Ariane Lewis, MD, and Steven Galetta, MD
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Reader Response: Opinion and Special Articles: Remote Evaluation of Acute Vertigo: Strategies and Technological Considerations

Carsten Tjell (Kristiansand, Norway)
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I fully agree with the paper by Green et al.[1] However, these strategies are not always easy for a young colleague in a busy emergency department. I have many years of positive experiences using the model presented in this short paper,[2] which describes an algorithm with 3 components: the Romberg test, the horizontal smooth pursuit eye movement test, and tragal pressure. This algorithm can be used at bedside without any equipment or previous training diagnosing nystagmus. I would propose for the head impulse test, nystagmus evaluation, and test of skew (HINTS) and my proposed algorithm to be used together. The tragal pressure is only relevant in the peripheral cases; I include it because of my clinical experience of overlooked cholesteatomas, presenting as a small amount of cerumen at the top of the tympanic membrane.

1. Green KE, Pogson JM, Otero-Millan J, et al. Opinion and special articles: Remote evaluation of acute vertigo: strategies and technological Considerations. *Neurology*. 2021;96(1):34-38.
2. Tjell C. Sudden vertigo—fateful or just fearful? *Tidsskr Nor Laegeforen*. 2020;140(1). doi: 10.4045/tidsskr.19.0671.

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Author Response: Opinion and Special Articles: Remote Evaluation of Acute Vertigo Strategies and Technological Considerations

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We appreciate the comments by Dr. Tjell on our editorial.[1] To our knowledge, the diagnostic accuracy of the proposed tests has not been compared with the head impulse test, nystagmus evaluation, and test of skew (HINTS) battery. There is evidence from Carmona et al.[2] regarding the diagnostic value of truncal ataxia, but the same study found that the HINTS battery was far more sensitive and specific in distinguishing central from peripheral vestibulopathy. The evaluation of nystagmus only by its disruptive effect on smooth pursuit can lead to clinical errors because the pursuit performance requires visual attention, declines with age, can be influenced by a spontaneous nystagmus, and is particularly susceptible to the influence of medications. Although tragal compression, that is, the Hennebert test, can elicit vestibular nystagmus in some patients by changing the middle ear pressure, it is uncertain whether it might have a diagnostic value in the setting of acute vestibular syndrome. Finally, future automated predictive models combining the proposed test battery (excluding tragal compression) with HINTS may prove to be beneficial for remote screening of acute dizzy patients. However, a very large multicenter international collaboration would be required for such a study to create clinically impactful results.

1. Green KE, Pogson JM, Otero-Millan J, et al. Opinion and special articles: Remote evaluation of acute vertigo: strategies and technological Considerations. *Neurology*. 2021;96(1):34-38.
2. Carmona S, Martínez C, Zalazar G, et al. The diagnostic accuracy of truncal ataxia and HINTS as cardinal signs for acute vestibular syndrome. *Front Neurol*. 2016;7:125.

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Editors' Note: Association of Group A Streptococcus Exposure and Exacerbations of Chronic Tic Disorders: A Multinational Prospective Cohort Study

In "Association of Group A Streptococcus Exposure and Exacerbations of Chronic Tic Disorders: A Multinational Prospective Cohort Study," Martino et al. followed 715 children with chronic tic disorders (CTD) for up to 16 months and found that there was no significant association between group A streptococcus (GAS) pharyngeal exposures and tic exacerbations, although group A streptococcus exposure was associated with exacerbation of ADHD symptoms. On behalf of the European Immuno-Neuropsychiatric Association Scientific and Medical Advisory Group, Ubhi et al. praised the study but pointed out that because most patients had a previous diagnosis of Tourette syndrome, making it unlikely their tics were immune-mediated, these findings should not be used to refute the existence of immune-mediated tics or extrapolated to conclude immune-mediated tics are not associated with GAS exposure. Martino agreed that CTD are distinct from pediatric autoimmune neuropsychiatric disorders associated with streptococcal infections (PANS/PANDAS) but noted there is growing evidence that CTD may be the result of immune-mediated mechanisms triggered by genetics and environmental exposures. Martino et al. and Ubhi et al. suggested there is a need for additional investigation into the relationship between GAS exposure and ADHD symptoms.

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Reader Response: Association of Group A Streptococcus Exposure and Exacerbations of Chronic Tic Disorders: A Multinational Prospective Cohort Study

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The European Immuno-Neuropsychiatric Association Scientific and Medical Advisory Group welcomes this study that considered associations between exposure to group A streptococcus (GAS) and chronic tic disorders.[1] This well-conducted study, with a balanced discussion, found no significant association between GAS and exacerbation of tics in children with chronic tic disorders. It is important that the study population was composed mainly of patients with a previous diagnosis of Tourette syndrome (90.8%). Thus, it is likely that the vast majority did not meet the diagnostic criteria for immune-mediated conditions, such as pediatric acute-onset neuropsychiatric syndrome/pediatric autoimmune neuropsychiatric disorders associated with streptococcal infections (PANS/PANDAS), which is differentiated from Tourette syndrome by presenting with an acute onset of several neuropsychiatric symptoms that may include tics or movement disorders. The results and conclusions should not therefore be used to refute the existence of immune-mediated conditions, including PANS/PANDAS, or to imply that these conditions do not similarly have an association with GAS infection. It is also interesting that an association between exacerbation of attention-deficit/hyperactivity disorder symptoms and GAS exposure was found, which we agree should be further investigated. The value of looking

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for GAS immune-mediated changes in the context of acute-onset tic disorders is not delineated by this study, and we feel that point should be highlighted.

1. Martino D, Schrag A, Anastasiou Z, et al. Association of group A streptococcus exposure and exacerbations of chronic tic disorders: a multinational prospective cohort study. *Neurology*. 2021;96(12):e1680-e1693.

Author Response: Association of Group A Streptococcus Exposure and Exacerbations of Chronic Tic Disorders: A Multinational Prospective Cohort Study

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We are grateful to the European Immuno-Neuropsychiatric Association Scientific and Medical Advisory Group for their interest and positive comments on our study. As correctly pointed out, our study exclusively enrolled patients with an established diagnosis of chronic tic disorders, the vast majority of whom had Tourette syndrome.[1] Pediatric acute-onset neuropsychiatric syndrome/pediatric autoimmune neuropsychiatric disorders associated with streptococcal infections (PANS/PANDAS) represent a clinical spectrum that is substantially different from the clinical course and phenotype of chronic tic disorders.[2] There is growing evidence supporting a direct role of immune-mediated mechanisms in the pathogenesis of chronic, neurodevelopmental disorders such as Tourette syndrome, obsessive-compulsive disorder, and ADHD.[3] Genetic and early environmental exposures might prompt a very early priming of neural and immune development, which might lead to altered trajectories of maturation and, ultimately, to behavioral abnormalities and hyperactive immune-inflammatory responses.[3,4] Our study focused on the role of a specific exogenous trigger (group A streptococcus) in the temporal course of core behavioral features of chronic tic disorders during early adolescence. Our results do not refute a mechanistic link between immune and neural mechanisms during early development in chronic tic disorders. At the same time, we agree that our clinical overall message applies to chronic tic disorders but not to acute-onset behavioral presentations which encompass PANS/PANDAS.

1. Martino D, Schrag A, Anastasiou Z, et al. Association of group A *Streptococcus* exposure and exacerbations of chronic tic disorders: a multinational prospective Cohort study. *Neurology*. 2021;96(12):e1680-e1693.
2. Chang K, Frankovich J, Cooperstock M, et al. Clinical evaluation of youth with pediatric acute-onset neuropsychiatric syndrome (PANS): recommendations from the 2013 PANS Consensus Conference. *J Child Adolesc Psychopharmacol*. 2015;25(1):3-13.
3. Martino D, Johnson I, Leckman JF. What does immunology have to do with normal brain development and the pathophysiology underlying Tourette syndrome and related neuropsychiatric disorders? *Front Neurol*. 2020;11:567407.
4. Frick L, Pittenger C. Microglial dysregulation in OCD, Tourette syndrome, and PANDAS. *J Immunol Res*. 2016;2016:8606057.

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